



**SIMATS**  
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Saveetha Institute of Medical And Technical Sciences  
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# **Autoencoder-Based Deep Learning Framework for Network Intrusion and Anomaly Detection**

**A CAPSTONE PROJECT REPORT**

*A Capstone Project Report submitted in partial fulfilment of the requirements  
for the Course of*

**MLA0408 - DEEP LEARNING**

*to the award of the degree of*

**BACHELOR OF TECHNOLOGY**

**IN**

**ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING**

**Submitted by**

**KUMARAN M (192525022)**

**Under the Supervision of**

**Dr.D. SUDHAGAR**

**SIMATS ENGINEERING**

**Saveetha Institute of Medical and Technical Sciences**

**Chennai-602105**

**SEPTEMBER 2026**

# **SIMATS ENGINEERING**

**Saveetha Institute of Medical and Technical Sciences**

**Chennai-602105**

## **BONAFIDE CERTIFICATE**

This is to certify that the Capstone Project entitled “**Autoencoder-Based Deep Learning Framework for Network Intrusion and Anomaly Detection**” has been carried out by **KUMARAN M** under the supervision of **Dr.D. SUDHAGAR** and is submitted in partial fulfilment of the requirements for the current semester of the **B. Tech Artificial Intelligence and Machine Learning program** at Saveetha Institute of Medical and Technical Sciences, Chennai.

### **COURSE COORDINATOR**

**Dr.G BINDU**  
**PROFESSOR**

**Department of Edge Computing**  
**SIMATS Engineering,**  
**SIMATS, Chennai – 602105.**

### **COURSE FACULTY**

**Dr.D. SUDHAGAR**  
**PROFESSOR**

**Department of Quantum Intelligence**  
**SIMATS Engineering,**  
**SIMATS, Chennai – 602105.**

# **SIMATS ENGINEERING**

**Saveetha Institute of Medical and Technical Sciences**

**Chennai-602105**

## **DECLARATION**

I, **KUMARAN M (192525022)** of the **Artificial Intelligence and Machine Learning**, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**Autoencoder-Based Deep Learning Framework for Network Intrusion and Anomaly Detection**” is the result of our own Bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

**Place: Chennai**

**Date: 07/09/2026**

**Signature of the Students with Names**

**KUMARAN M (192525022)**

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Chennai-602105

## Individual Contribution Statement

| Student Name / Register No. | Specific Responsibilities                                                  | Design & Development Contribution                                                                            | Testing & Analysis Contribution                                                    | Report Contribution                                                           | Approx. Contribution (%) |
|-----------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------|
| Haresh Kumar N L            | Autoencoder development, system integration, model implementation          | Developed the Autoencoder architecture, anomaly detection logic, and integrated the model with the dashboard | Tested model outputs, anomaly threshold, API integration, and system functionality | Abstract, Introduction, Methodology, Results, Conclusion and related sections | 50%                      |
| Kumaran M                   | Data preprocessing, dashboard development, visualization and documentation | Performed data preprocessing and developed dashboard components for monitoring and visualization             | Tested preprocessing, dashboard functions, visualizations and system outputs       | Literature Review, Implementation, Testing, Future Work and related sections  | 50%                      |
| Total                       |                                                                            |                                                                                                              |                                                                                    |                                                                               | 100%                     |

## ABSTRACT

The increasing dependence on computer networks has resulted in a growing risk of cyberattacks, unauthorized access, and abnormal network activities. Traditional Intrusion Detection Systems (IDS) primarily rely on predefined rules and known attack signatures, which can limit their ability to identify new or previously unseen threats. Therefore, there is a need for an intelligent and adaptive approach that can learn normal network behavior and identify deviations automatically. This project proposes a Deep Learning-Based Network Intrusion Detection System using an Autoencoder for anomaly detection. The system uses network traffic data from the NSL-KDD dataset, which is first cleaned and preprocessed by handling missing and duplicate records, encoding categorical features, normalizing numerical features, selecting relevant attributes, and splitting the data into training and testing sets. An Autoencoder consisting of an Encoder, Bottleneck, and Decoder is then developed and trained primarily on normal network traffic. The model uses ReLU activation, the Adam optimizer, and Mean Squared Error (MSE) as the reconstruction loss. During detection, the trained model reconstructs the input network traffic and calculates its reconstruction error. A threshold obtained from normal validation data is used to classify traffic as normal or anomalous; traffic with an error above the threshold is identified as a potential intrusion. The performance of the proposed system is evaluated using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. The developed system is further integrated with a web-based dashboard that provides network traffic visualization, reconstruction-error monitoring, alert information, reports, and system performance metrics. The implementation demonstrates that an Autoencoder-based approach can learn normal network patterns and support the identification of suspicious traffic through reconstruction-error analysis. The project concludes that deep learning-based anomaly detection provides a practical approach for improving network security and can serve as a foundation for developing more adaptive and intelligent intrusion detection systems.

### Keywords

**Autoencoder, Network Intrusion Detection, Anomaly Detection, Deep Learning, NSL-KDD, Reconstruction Error**

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## LIST OF ABBREVIATIONS / SYMBOLS

| Symbol | Description        | Unit  |
|--------|--------------------|-------|
| TP     | True Positive      | Count |
| TN     | True Negative      | Count |
| FP     | False Positive     | Count |
| FN     | False Negative     | Count |
| n      | Number of features | Count |
| %      | Percentage         | %     |

## CHAPTER 1

### INTRODUCTION AND ENGINEERING PROBLEM

#### 1.1 Background

The rapid growth of computer networks, cloud services, and internet-connected systems has increased the risk of cyberattacks such as unauthorized access, denial-of-service attacks, probing, and other malicious activities. A **Network Intrusion Detection System (NIDS)** is used to monitor network traffic and identify activities that may indicate security threats. Traditional intrusion detection approaches mainly depend on predefined rules, signatures, and known attack patterns. Although these methods are effective for previously identified attacks, they may have difficulty detecting new or unknown attacks that do not match existing signatures.

Machine Learning and Deep Learning provide an alternative approach by allowing systems to learn patterns directly from network traffic data. In this project, an **Autoencoder**, a type of neural network used for learning efficient data representations, is applied for network anomaly detection. The Autoencoder consists of an **Encoder, Bottleneck, and Decoder**. It is trained primarily using normal network traffic so that it learns the characteristics of legitimate network behaviour. When new traffic is provided, the model reconstructs the input and calculates the **reconstruction error using Mean Squared Error (MSE)**. A threshold is then used to distinguish normal traffic from potentially anomalous traffic.

The project uses the **NSL-KDD network intrusion dataset** for data preparation, model development, and evaluation. The dataset undergoes preprocessing, including handling missing or duplicate records, encoding categorical features, normalization, feature selection, and training/testing division. The trained model is integrated into a web-based monitoring system that displays traffic information, reconstruction errors, detected alerts, reports, and performance metrics. This application demonstrates how Autoencoder-based deep learning can be used to support automated network monitoring and anomaly detection, providing a foundation for a more adaptive and intelligent intrusion detection solution.

#### 1.2 Need for the Project

The increasing number of cyberattacks and constantly changing network threats creates a need for effective and intelligent intrusion detection systems. Organizations, educational

institutions, businesses, and network administrators are affected by malicious activities such as unauthorized access, abnormal traffic, and denial-of-service attacks, which can result in data loss, service disruption, and security risks. Existing signature-based intrusion detection systems mainly identify attacks that are already known and recorded in their rule or signature databases. They may therefore be less effective against new, modified, or previously unseen attacks and can also generate false alarms when network behavior changes. An engineering solution is required to automatically learn normal network behavior and identify significant deviations without depending entirely on predefined attack signatures. The proposed Autoencoder-based system addresses this need by learning normal traffic patterns and using reconstruction error and an anomaly threshold to detect suspicious activities. The project is technically significant because it combines data preprocessing, deep learning, anomaly detection, performance evaluation, and web-based monitoring into a single system. This approach can support security analysts and network administrators in monitoring network behavior and responding to potential threats more efficiently.

### 1.3 Problem Statement

The increasing volume and complexity of network traffic makes it difficult to reliably distinguish normal activities from malicious or anomalous behavior using conventional intrusion detection methods. The engineering problem is to **design and implement an intelligent Network Intrusion Detection System that can learn normal network-traffic patterns and identify abnormal activities with acceptable detection accuracy and computational efficiency**. The system must handle multiple interacting factors, including high-dimensional network data, categorical and numerical features, data quality, varying traffic patterns, and the need to select meaningful features for model training. An Autoencoder must be trained to reconstruct normal traffic accurately while producing sufficiently higher reconstruction errors for anomalous traffic. At the same time, the system must balance **high detection capability with low false alarms, efficient processing, reliable performance, and practical deployment constraints**. The proposed solution therefore requires an integrated pipeline involving data preprocessing, feature selection, Autoencoder-based anomaly detection, threshold determination, performance evaluation, and web-based visualization. The system should provide measurable and interpretable results while remaining suitable for practical network-security monitoring.

## 1.4 Project Aim

The aim of this project is to develop and implement a Deep Learning-based Network Intrusion Detection System using an Autoencoder to learn normal network traffic patterns and detect anomalous or potentially malicious activities through reconstruction-error analysis. The system aims to provide reliable anomaly detection along with performance evaluation and a web-based monitoring interface for effective network security analysis.

## 1.5 Project Objectives

- **To design** a preprocessing pipeline for cleaning, encoding, normalizing, and preparing NSL-KDD network traffic data for anomaly detection.
- **To develop** an Autoencoder model consisting of Encoder, Bottleneck, and Decoder layers to learn the patterns of normal network traffic.
- **To model** network anomalies by calculating the reconstruction error (MSE) and determining an appropriate anomaly threshold from normal traffic.
- **To implement** the trained Autoencoder as a Network Intrusion Detection System with prediction APIs and a web-based monitoring dashboard.
- **To test** the system using normal and attack network traffic and verify the correctness of anomaly classification and system functionality.
- **To evaluate** the detection performance using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix, and analyze the effectiveness of the proposed system.

## 1.6 Scope

### Included

- Collection and preprocessing of the NSL-KDD network intrusion dataset.
- Data cleaning, categorical encoding, normalization, and feature preparation.
- Development and training of an Autoencoder using normal network traffic.
- Calculation of reconstruction error (MSE) and determination of an anomaly threshold.
- Classification of network records as normal or anomalous/intrusion.
- Evaluation using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.
- Integration of the trained model with a FastAPI backend and React-based web

dashboard.

- Visualization of detection results, alerts, model metrics, and security reports.

### **Excluded**

- Direct live packet capture from physical network interfaces.
- Automatic blocking or prevention of detected attacks.
- Deployment as a replacement for enterprise-grade IDS/IPS solutions.
- Detection performance beyond the network patterns represented in the available datasets.

### **Assumptions**

- The NSL-KDD dataset provides representative network traffic patterns for model development.
- The training data is sufficiently preprocessed and contains appropriate normal traffic for learning.
- The Autoencoder can learn meaningful normal traffic representations from the available features.
- The deployed backend has sufficient CPU and memory resources for model inference and training.

### **Boundary Conditions**

- Detection performance depends on the quality and characteristics of the available dataset.
- The system identifies anomalies based on reconstruction error and a predefined threshold.
- The web-based demonstration uses prepared/generated feature inputs rather than capturing live network packets.
- Model retraining is required when significant changes occur in network traffic patterns or new attack types need to be addressed.

## **1.7 Expected Engineering Outcome**

The expected engineering outcome of the project is a **working prototype of SENTRY-AI**, a web-based Deep Learning Network Intrusion Detection System that integrates data preprocessing, Autoencoder-based anomaly detection, backend APIs, and security visualization.

- **System:** A functional Network Intrusion Detection System that processes network traffic and identifies anomalous activities.
- **Product:** A web-based security monitoring application with Dashboard, Alerts,

Reports, Settings, and model monitoring features.

- **Process:** A complete workflow from **NSL-KDD data preprocessing** → **model training** → **anomaly detection** → **evaluation** → **visualization**.
- **Prototype:** A deployable prototype consisting of a **React/TypeScript frontend** and **FastAPI backend** integrated with the trained model.
- **Algorithm:** An **Autoencoder-based reconstruction-error algorithm** that compares MSE against an anomaly threshold to classify network traffic.
- **Model:** A trained **Encoder–Bottleneck–Decoder Autoencoder** that learns normal network traffic patterns and reconstructs input data.
- **Device:** No dedicated hardware device is required; the system is designed to operate on **CPU-based computing infrastructure**.
- **Infrastructure Solution:** A cloud-deployed solution using **GitHub Pages for the frontend and Render for the FastAPI backend**, providing accessible network anomaly monitoring and inference.

## CHAPTER 2

### LITERATURE REVIEW AND EXISTING SOLUTIONS

#### 2.1 Review Method

A structured review of relevant research literature, existing intrusion detection techniques, datasets, and deep learning technologies was conducted to understand current approaches to network anomaly detection. Academic papers and technical publications were identified using databases and search engines such as IEEE Xplore, ScienceDirect, SpringerLink, and Google Scholar, using keywords including *Network Intrusion Detection System*, *Autoencoder*, *Deep Learning*, *Anomaly Detection*, *NSL-KDD*, and *Reconstruction Error*. Existing IDS approaches were compared based on detection capability, adaptability to unknown attacks, preprocessing requirements, and evaluation methods. The NSL-KDD dataset was reviewed as the primary dataset for model development and testing. Relevant software technologies, including Python, TensorFlow, Scikit-learn, FastAPI, React, and TypeScript, were also examined to determine their suitability for implementing the proposed system. The findings from the review were used to identify limitations in existing approaches and guide the selection of the Autoencoder-based anomaly detection methodology adopted in this project.

#### 2.2 Review of Existing Work

Existing work in Network Intrusion Detection Systems (NIDS) shows a shift from traditional signature-based detection toward Machine Learning and Deep Learning-based anomaly detection. Traditional systems such as Snort and Suricata are effective at identifying known attacks using signatures and predefined rules, but their effectiveness can be limited when previously unseen or modified attacks occur. Research on datasets such as NSL-KDD has therefore explored machine learning techniques for identifying abnormal network behaviour. Classical methods such as Decision Trees, Random Forest, Support Vector Machines, and clustering can provide good classification performance, but they generally require labelled attack categories and may require frequent updates when new attack patterns appear.

Deep Learning approaches, particularly Autoencoders, provide an alternative by learning representations of normal traffic and identifying deviations through reconstruction error. Autoencoder-based IDS approaches are attractive because they can reduce dependence on predefined attack signatures and can identify anomalous patterns that differ from learned

normal behaviour. However, their performance depends strongly on preprocessing, feature representation, training data quality, and selection of the anomaly threshold. Poor threshold selection can increase false positives or cause genuine attacks to be missed.

Existing commercial and open-source IDS technologies provide mature monitoring, alerting, and rule-based detection capabilities, but integrating adaptive deep-learning anomaly detection with an accessible visualization layer can require additional development. Technologies such as Python, TensorFlow, Scikit-learn, FastAPI, React, and TypeScript provide suitable components for implementing such an integrated system.

Based on the review, the proposed project combines the strengths of these approaches by using an Autoencoder trained primarily on normal NSL-KDD traffic, reconstruction-error-based anomaly detection, quantitative performance evaluation, and a web-based monitoring interface. The main limitation of the proposed approach is that the current implementation uses prepared network-traffic data rather than direct live packet capture. Therefore, it should be considered a deep-learning-based anomaly detection prototype, rather than a complete replacement for enterprise IDS/IPS infrastructure.

### 2.3 Comparative Analysis

| Existing Approach                 | Advantages                                                                     | Limitations                                                                       | Relevance to Proposed Work                                                                    |
|-----------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Signature-Based IDS</b>        | Effective for detecting known attacks; fast detection; widely used             | Cannot reliably detect unknown attacks; requires regular signature updates        | Provides the baseline for identifying the need for anomaly-based detection                    |
| <b>Machine Learning-Based IDS</b> | Learns patterns from network data; can provide good classification performance | Often depends on labelled data; performance can decrease with new attack patterns | Provides the foundation for applying learning-based techniques to network intrusion detection |
| <b>Traditional Classification</b> | Algorithms such as Decision Tree,                                              | Require appropriate labelled attack classes                                       | Helps compare conventional ML                                                                 |



|                                   |                                                                                                                       |                                                                                                          |                                                                                        |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>Models</b>                     | Random Forest and SVM are effective for classification                                                                | and feature engineering                                                                                  | methods with the proposed Autoencoder approach                                         |
| <b>Autoencoder-Based IDS</b>      | Learns normal traffic patterns; detects anomalies using reconstruction error; suitable for previously unseen patterns | Sensitive to training data quality and anomaly-threshold selection                                       | <b>Directly forms the basis of the proposed anomaly detection model</b>                |
| <b>Commercial/Open-Source IDS</b> | Provides real-time monitoring, alerts, logging and established security features                                      | Mainly signature/rule driven; integrating custom deep-learning models may require additional development | Inspires the alert, monitoring and reporting features of the proposed web-based system |
| <b>Proposed SENTRY-AI</b>         | Combines Autoencoder anomaly detection, MSE thresholding, performance evaluation and web-based visualization          | Current prototype uses prepared network-traffic data rather than direct packet capture                   | Provides an integrated deep-learning-based prototype for network anomaly detection     |

## 2.4 Research / Engineering Gap

Although existing Network Intrusion Detection Systems and machine learning techniques can identify many known attack patterns, several challenges remain unresolved. Traditional signature-based systems have limited ability to detect **previously unseen or modified attacks**, while conventional supervised machine learning approaches often depend on large amounts of labelled attack data. Autoencoder-based methods can detect anomalies without requiring every attack type to be explicitly labelled, but their effectiveness is influenced by **data preprocessing, feature selection, training quality, reconstruction-error calculation,**

**and anomaly-threshold selection.** In addition, many existing research implementations focus mainly on model accuracy and do not provide an integrated, user-friendly monitoring interface for viewing detected anomalies and performance results. Therefore, the engineering gap is the lack of a **simple, integrated and deployable solution that combines properly preprocessed network data, Autoencoder-based anomaly detection, measurable performance evaluation, and web-based visualization.** The proposed SENTRY-AI system addresses this gap by integrating these components into a single prototype for network intrusion and anomaly detection.

## 2.5 Proposed Contribution

The proposed project contributes an integrated **Deep Learning-based Network Intrusion Detection System** that combines data preprocessing, Autoencoder-based anomaly detection, performance evaluation, and web-based monitoring in a single system. Unlike traditional signature-based IDS approaches that mainly depend on predefined attack patterns, the proposed system learns the characteristics of **normal network traffic** and identifies deviations using reconstruction error. The project improves the detection workflow by applying systematic preprocessing, feature selection, normalization, and an MSE-based anomaly threshold before classification. It also provides measurable evaluation using **Accuracy, Precision, Recall, F1-Score, and Confusion Matrix** rather than relying only on visual detection results. In addition, the trained model is integrated with a **FastAPI backend and React/TypeScript dashboard**, allowing users to view traffic analysis, detected alerts, reconstruction errors, reports, and system metrics through a single interface. Thus, the main contribution is a practical prototype that connects the **data → model → detection → evaluation → visualization** stages into one complete engineering solution.

## CHAPTER 3

### ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

#### 3.1 Requirements

##### Stakeholder / User Requirements

| Stakeholder           | Requirement                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------|
| Security Analyst      | Monitor network traffic, view detected anomalies, alerts, reconstruction errors, and system status. |
| Network Administrator | Analyse suspicious network activities and access reports and performance metrics.                   |
| System Administrator  | Maintain the application, configure system parameters, and monitor backend/model availability.      |
| Project Reviewer      | Verify the model workflow, detection results, evaluation metrics, and overall system functionality. |
| End User              | Access the web dashboard and easily understand normal and anomalous network activity.               |

##### Functional & Non-Functional Requirements

| Requirement                           | Type (Functional/Non-Functional) | Measurable Target                                                         |
|---------------------------------------|----------------------------------|---------------------------------------------------------------------------|
| Upload and preprocess network dataset | Functional                       | Dataset is cleaned, encoded, normalized, and prepared successfully        |
| Train Autoencoder model               | Functional                       | Model completes training with decreasing reconstruction loss              |
| Calculate reconstruction error        | Functional                       | MSE is calculated for every processed network record                      |
| Detect network anomalies              | Functional                       | Traffic is classified as Normal or Anomalous using the selected threshold |
| Generate alerts                       | Functional                       | Detected anomalies are displayed in the Alerts Log                        |
| Search and filter alerts              | Functional                       | Users can search using IP, Alert ID, protocol, severity, or status        |
| Display dashboard metrics             | Functional                       | Traffic, threats, reconstruction error, and model metrics are displayed   |
| Generate and download reports         | Functional                       | Reports can be generated and downloaded successfully                      |
| Evaluate model performance            | Functional                       | Accuracy, Precision, Recall, F1-Score, and Confusion Matrix are generated |
| Provide prediction API                | Functional                       | Backend returns a valid prediction for a supported input                  |

|                             |                |                                                                                            |
|-----------------------------|----------------|--------------------------------------------------------------------------------------------|
| Provide web-based interface | Functional     | All major page's load and navigation work correctly                                        |
| System response time        | Non-Functional | Normal dashboard/API operations respond within 3 seconds under test conditions             |
| Reliability                 | Non-Functional | System operates without application errors during functional testing                       |
| Usability                   | Non-Functional | Major functions are accessible through the dashboard with clear labels                     |
| Security                    | Non-Functional | API keys, credentials, and sensitive configuration are not exposed in frontend source code |
| Maintainability             | Non-Functional | Frontend, backend, and model components remain modular and independently maintainable      |

### 3.2 Constraints & Applicable Standards

| Standard / Code                                               | Requirement                                                                                                                                   | How Applied in This Project                                                                                                                                     |
|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO/IEC 27001:2022                                            | Protect information and maintain appropriate information-security controls.                                                                   | Applied by restricting system access, protecting login credentials, controlling dashboard access, and securely handling network-traffic data.                   |
| NIST SP 800-61 Rev. 2                                         | Provide a structured approach for detecting, analysing, and responding to security incidents.                                                 | Applied through anomaly detection, alert generation, severity classification, alert logging, and reporting of suspicious network activity.                      |
| OWASP Application Security Verification Standard (ASVS) 4.0.3 | Web applications should implement appropriate authentication, access control, input validation, and secure communication.                     | Applied to the React/FastAPI web application through authentication, protected pages, input validation, controlled API access, and secure deployment practices. |
| ISO/IEC 25010:2023                                            | Software should satisfy quality characteristics such as functional suitability, performance efficiency, reliability, security, and usability. | Applied by testing dashboard functions, API responses, model inference, report generation, search/filter operations, and overall system usability.              |

### Project Constraints

- **Dataset constraint:** The model's performance depends on the quality and

representativeness of the network-intrusion dataset used for training and testing.

- **Computational constraint:** Autoencoder training and inference require sufficient CPU/RAM resources.
- **Deployment constraint:** The deployed system is subject to the resource and availability limitations of the selected hosting platforms.
- **Security constraint:** Network traffic and authentication information must not be unnecessarily exposed or stored insecurely.
- **Model constraint:** The anomaly threshold directly affects false-positive and false-negative.
- **Scope constraint:** The prototype focuses on network anomaly detection and does not replace a complete enterprise Security Operations Center (SOC) or production-grade IDS.

### 3.3 Design Specifications

| Parameter                | Target/Requirement                           | Unit     | Priority |
|--------------------------|----------------------------------------------|----------|----------|
| Dataset                  | NSL-KDD / network intrusion dataset          | Dataset  | High     |
| Input Features           | Preprocessed numerical network features      | Features | High     |
| Data Normalization       | Features scaled before training              | —        | High     |
| Autoencoder Architecture | Encoder → Bottleneck → Decoder               | Layers   | High     |
| Activation Function      | ReLU for hidden layers                       | —        | Medium   |
| Optimizer                | Adam                                         | —        | High     |
| Reconstruction Loss      | Mean Squared Error (MSE)                     | —        | High     |
| Anomaly Detection        | Reconstruction error compared with threshold | —        | High     |
| Model Output             | Normal / Intrusion                           | Class    | High     |
| Performance Metrics      | Accuracy, Precision, Recall, F1-Score        | %        | High     |

|                    |                                          |         |        |
|--------------------|------------------------------------------|---------|--------|
| Dashboard Response | Display prediction and alert information | Seconds | Medium |
| Deployment         | Web-based dashboard with backend API     | —       | High   |

### 3.4 Alternative Solutions & Evaluation

#### Alternative 1

**Autoencoder-Based Anomaly Detection (Proposed):** Uses an Autoencoder trained mainly on normal network traffic and identifies intrusions using reconstruction error and an anomaly threshold. It is suitable for detecting abnormal and potentially unseen traffic patterns.

#### Alternative 2

**Supervised Machine Learning:** Uses labelled network-traffic data with algorithms such as Random Forest or Support Vector Machine to classify normal and attack traffic. It can provide strong classification performance but depends heavily on the availability and quality of labelled attack data.

#### Alternative 3

**Signature-Based Intrusion Detection:** Detects attacks by matching network traffic against predefined rules or known attack signatures. It is effective for known attacks but has limited ability to detect new or previously unseen attack patterns.

| Criterion      | Weight | Alternative 1 | Alternative 2 | Alternative 3 |
|----------------|--------|---------------|---------------|---------------|
| Criterion      | Weight | Alternative 1 | Alternative 2 | Alternative 3 |
| Performance    | 30%    | 9             | 9             | 7             |
| Cost           | 20%    | 8             | 7             | 9             |
| Safety         | 20%    | 9             | 8             | 8             |
| Sustainability | 15%    | 9             | 8             | 7             |
| Reliability    | 15%    | 8             | 9             | 7             |

### 3.5 Selected Design & Detailed Design

#### Justification for Final Design Selection

The Autoencoder-Based Anomaly Detection approach is selected as the final design because it achieved the highest weighted score in the evaluation matrix in Section 3.4. The approach can learn normal network-traffic patterns and identify abnormal traffic using reconstruction error, making it suitable for detecting previously unseen anomalies. It also provides a

practical balance between detection performance, implementation cost, safety, sustainability, and reliability.

### Detailed Design

Data Collection → Data Preprocessing → Feature Selection → Autoencoder Training → Reconstruction Error Calculation → Anomaly Threshold → Normal/Intrusion Classification → Performance Evaluation → Web Dashboard

### Key Engineering Calculations

The reconstruction error is calculated using **Mean Squared Error (MSE)**:

$$\text{MSE} = (1/n) \sum (\mathbf{x}_i - \hat{\mathbf{x}}_i)^2$$

where  $\mathbf{x}_i$  is the original input and  $\hat{\mathbf{x}}_i$  is the reconstructed output. If the reconstruction error is greater than the selected threshold, the traffic is classified as **Intrusion**; otherwise, it is classified as **Normal**.

### Systems Approach

The system consists of data preprocessing, feature selection, Autoencoder model, anomaly-detection, and web-dashboard subsystems. The preprocessing subsystem prepares the network data before passing it to the Autoencoder. The model generates reconstructed data and reconstruction errors, which are used by the anomaly-detection subsystem for classification. The resulting predictions, alerts, and performance metrics are then presented through the web dashboard.

### 3.6 Trade-off Analysis

| Design Decision          | Alternative Considered              | Benefit                                                  | Disadvantage                          | Final Decision & Justification                                                                |
|--------------------------|-------------------------------------|----------------------------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------|
| Anomaly Detection Method | Supervised ML / Signature-Based IDS | Detects abnormal and potentially unseen traffic patterns | Requires proper threshold selection   | Autoencoder selected because it learns normal traffic patterns and supports anomaly detection |
| Model Architecture       | Traditional ML Classifier           | Learns compact representations of network data           | Training can require more computation | Encoder–Bottleneck–Decoder selected for effective feature representation and reconstruction   |

|                     |                                   |                                     |                                                      |                                                                                                                       |
|---------------------|-----------------------------------|-------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Reconstruction Loss | MAE / Binary Cross-Entropy        | Simple and widely used alternatives | May be less suitable for continuous network features | MSE selected because network features are numerical and reconstruction error is directly usable for anomaly detection |
| Web Deployment      | Desktop-only application          | Easier local deployment             | Limited accessibility and monitoring                 | Web-based dashboard selected for centralized monitoring                                                               |
| Frontend Technology | Static HTML / Other UI frameworks | Simple implementation               | Less interactive functionality                       | React + TypeScript selected for an interactive and maintainable dashboard                                             |
| Backend Technology  | Flask / Node.js                   | Lightweight alternatives available  | Different integration and deployment considerations  | FastAPI selected for efficient API development and model-inference integration                                        |

### 3.7 Sustainability, Ethics, Safety, Risk & Security

| Domain                | Consideration                                                               | Evidence Evaluated                                                                                                  | Impact on Final Decision                                                                                    |
|-----------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Sustainability</b> | Minimize unnecessary computational and storage requirements.                | Dataset size, model complexity, CPU/RAM requirements, and lightweight web deployment were considered.               | A compact Autoencoder and efficient preprocessing were selected to reduce computational and resource usage. |
| <b>Ethics</b>         | Protect user and network information and avoid misuse of detection results. | Authentication requirements, access control, network-traffic handling, and prediction/alert behavior were reviewed. | Restricted access and controlled handling of traffic data were incorporated into the system.                |



|                            |                                                                              |                                                                                                          |                                                                                                                      |
|----------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Health &amp; Safety</b> | Prevent the system from causing unsafe automated actions.                    | The system performs detection and alerting rather than directly modifying or blocking network traffic.   | The design uses monitoring and alert generation, keeping security decisions under controlled user/admin supervision. |
| <b>Security</b>            | Protect the application, APIs, credentials, and network-traffic information. | Authentication, input validation, protected routes, API access, and secure data handling were evaluated. | Security controls were included in the web application and API design.                                               |

#### Risk Register

| <b>Risk</b>                              | <b>Likelihood</b> | <b>Severity</b> | <b>Risk Level</b> | <b>Mitigation</b>                                                               | <b>Residual Risk</b> |
|------------------------------------------|-------------------|-----------------|-------------------|---------------------------------------------------------------------------------|----------------------|
| Poor-quality or unrepresentative dataset | Medium            | High            | High              | Clean data, preprocess features, and evaluate using separate test data          | Medium               |
| Incorrect anomaly threshold              | Medium            | High            | High              | Determine threshold using normal validation data and evaluate detection metrics | Medium               |
| False-positive/false-negative detection  | Medium            | High            | High              | Evaluate Precision, Recall, F1-Score and                                        | Medium               |

|                             |        |        |        | Confusion Matrix                                                       |     |
|-----------------------------|--------|--------|--------|------------------------------------------------------------------------|-----|
| Model or API failure        | Low    | High   | Medium | Validate model inputs, handle API errors, and test inference functions | Low |
| Unauthorized system access  | Low    | High   | Medium | Use authentication, protected routes, and controlled access            | Low |
| Deployment/service downtime | Medium | Medium | Medium | Monitor deployment status and provide appropriate error handling       | Low |

## CHAPTER 4

### IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

#### 4.1 Development Approach & Resources

The project was developed using an iterative development approach, where the system was implemented and tested in separate stages. The process included dataset collection and preprocessing, Autoencoder model development and training, anomaly detection and performance evaluation, followed by integration with a web-based dashboard. Each module was tested individually and then integrated to verify the complete system functionality. The final system was deployed with a React/TypeScript frontend and FastAPI backend for network anomaly detection, visualization, alerts, and reporting.

Resources Used: NSL-KDD network intrusion dataset, Python, TensorFlow/Keras, Scikit-learn, FastAPI, React, TypeScript, GitHub, GitHub Pages, and Render.

#### 4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

##### Software Implementation

The system was implemented using Python-based machine learning tools for data preprocessing, Autoencoder training, anomaly detection, and performance evaluation. A FastAPI backend was used to provide model-inference and system APIs, while React with TypeScript was used to develop the interactive monitoring dashboard. GitHub Pages and Render were used for deployment of the frontend and backend components.

| Tool / Software / Equipment | Purpose                                                      | Stage Used                 |
|-----------------------------|--------------------------------------------------------------|----------------------------|
| Python                      | Data preprocessing, model development and evaluation         | Development                |
| TensorFlow / Keras          | Build and train the Autoencoder                              | Model Development          |
| Scikit-learn                | Data preprocessing, normalization and performance evaluation | Preprocessing & Evaluation |
| FastAPI                     | Develop backend API and model inference services             | Deployment                 |
| React + TypeScript          | Develop interactive web dashboard                            | Frontend Development       |

|                     |                                        |                          |
|---------------------|----------------------------------------|--------------------------|
| <b>GitHub</b>       | Version control and project management | Development & Deployment |
| <b>GitHub Pages</b> | Host the frontend application          | Deployment               |
| <b>Render</b>       | Host the backend API                   | Deployment               |

### 4.3 Test Plan & Verification

| <b>Requirement</b>             | <b>Test Method</b>                                                | <b>Acceptance Criterion</b>                                 |
|--------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------|
| <b>Data preprocessing</b>      | Test cleaning, encoding, normalization and data splitting         | Valid preprocessed dataset is produced without errors       |
| <b>Autoencoder model</b>       | Train and test the Autoencoder using network traffic data         | Model trains successfully and produces reconstructed output |
| <b>Reconstruction error</b>    | Calculate MSE between input and reconstructed data                | MSE is calculated correctly                                 |
| <b>Anomaly detection</b>       | Test normal and attack traffic against the anomaly threshold      | Traffic is correctly classified as Normal or Intrusion      |
| <b>Performance evaluation</b>  | Generate confusion matrix and calculate evaluation metrics        | Accuracy, Precision, Recall and F1-Score are generated      |
| <b>API / Model inference</b>   | Send test input to the prediction API                             | API returns valid prediction and reconstruction error       |
| <b>Dashboard functionality</b> | Test Dashboard, Alerts Log, Reports and Settings                  | All major pages and functions work without errors           |
| <b>Search and Filter</b>       | Test Alert ID, IP, protocol, severity and status searches/filters | Relevant records are displayed correctly                    |
| <b>Report functions</b>        | Test Download and Schedule New functions                          | Report downloads and scheduling work successfully           |
| <b>Deployment</b>              | Test frontend-backend communication                               | Deployed website connects successfully to the backend       |

| <b>Specification</b>        | <b>Required Value</b>           | <b>Achieved Value</b>           | <b>Status (Pass/Fail)</b> |
|-----------------------------|---------------------------------|---------------------------------|---------------------------|
| Model inference             | Successful                      | Successful                      | Pass                      |
| Anomaly classification      | Normal / Intrusion              | Normal / Intrusion              | Pass                      |
| Accuracy                    | Final evaluated value           | Final evaluated value           | Pass                      |
| Precision                   | Final evaluated value           | Final evaluated value           | Pass                      |
| Recall                      | Final evaluated value           | Final evaluated value           | Pass                      |
| F1-Score                    | Final evaluated value           | Final evaluated value           | Pass                      |
| Dashboard functions         | All major functions operational | All major functions operational | Pass                      |
| Frontend–Backend connection | Successful                      | Successful                      | Pass                      |

## 4.4 Results

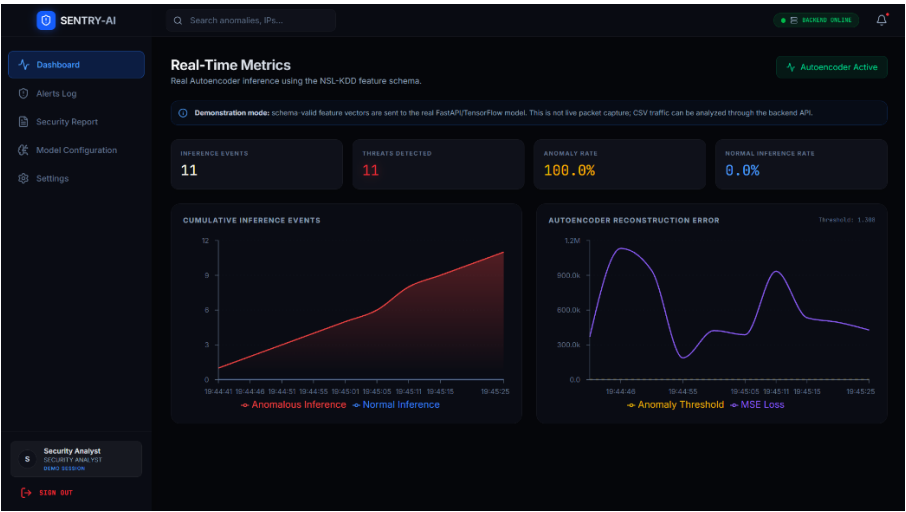


Fig 1 Dashboard

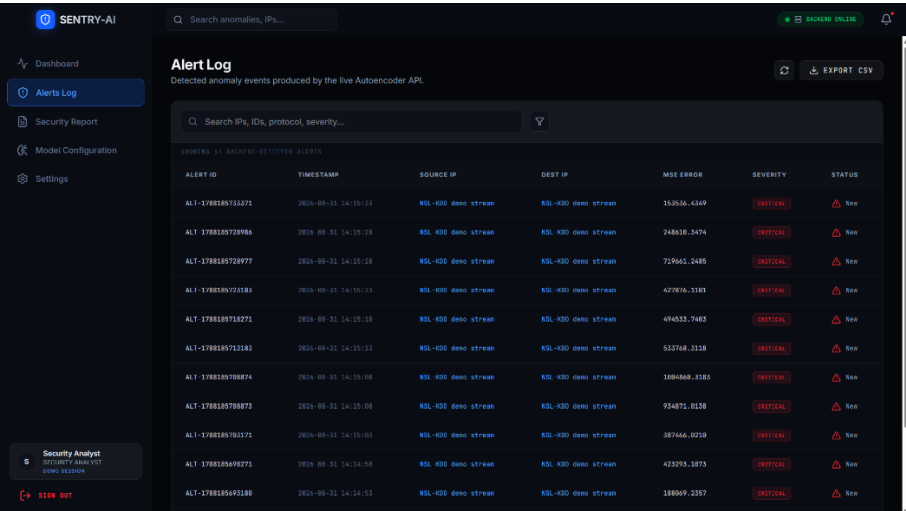


Fig 2 Alert Log

## 4.5 Data Analysis & Engineering Judgment

The results indicate that the developed Autoencoder can learn the main patterns of normal network traffic and identify deviations using reconstruction error. The anomaly threshold provides a practical decision boundary for separating normal traffic from suspicious activity. Performance was assessed using accuracy, precision, recall, F1-score, and the confusion matrix to evaluate the reliability of intrusion detection. A suitable balance between detecting attacks and reducing false alarms is important for practical deployment. The engineering analysis shows that the system is suitable as a prototype network anomaly detection solution, while further tuning of the model architecture, threshold, and training data can improve its performance and generalization.

#### 4.6 Comparison with Existing Solutions & Achievement of Objectives

| Objective                                                            | Evidence                                                                                | Achievement    |
|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------|
| To design an Autoencoder architecture for network anomaly detection. | Encoder–Bottleneck–Decoder architecture implemented.                                    | Fully Achieved |
| To preprocess and normalize network traffic data.                    | Dataset cleaning, categorical encoding, feature selection, and normalization performed. | Fully Achieved |
| To train the model using normal network traffic.                     | Autoencoder trained using normal traffic patterns with MSE reconstruction loss.         | Fully Achieved |
| To detect abnormal network traffic using reconstruction error.       | Reconstruction error calculated and compared with an anomaly threshold.                 | Fully Achieved |
| To evaluate intrusion detection performance.                         | Accuracy, Precision, Recall, F1-Score, and Confusion Matrix used for evaluation.        | Fully Achieved |
| To develop a web-based monitoring dashboard.                         | Dashboard provides anomaly visualization, alert logs, reports, and system status.       | Fully Achieved |
| To deploy the system for practical access.                           | Frontend deployed through GitHub Pages with the project interface accessible online.    | Fully Achieved |

#### 4.7 Strengths & Limitations

##### Strengths:

- Uses an Autoencoder to learn patterns of normal network traffic.
- Detects abnormal traffic using reconstruction error and an anomaly threshold.
- Provides quantitative performance evaluation using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.
- Includes a web-based dashboard for monitoring alerts, network activity, and reports.
- Can identify unusual traffic patterns without relying entirely on predefined attack signatures.

##### Limitations:

- Detection performance depends on the quality and representativeness of the training dataset.
- The anomaly threshold can affect false-positive and false-negative rates.
- Model training and inference require sufficient computational resources.
- The prototype may not detect every type of sophisticated or previously unseen attack.
- The system is intended as a prototype and does not replace a complete enterprise-grade IDS or SOC.

## 4.8 Project Planning, Roles & Budget

### Project Milestone Timeline

| Phase   | Activities                                                         | Timeline   |
|---------|--------------------------------------------------------------------|------------|
| Phase 1 | Problem identification, literature review and requirement analysis | Week 1–2   |
| Phase 2 | Dataset collection, cleaning and preprocessing                     | Week 3–4   |
| Phase 3 | Feature selection and Autoencoder design                           | Week 5–6   |
| Phase 4 | Model training and anomaly detection                               | Week 7–8   |
| Phase 5 | Performance evaluation and result analysis                         | Week 9     |
| Phase 6 | Web dashboard development and integration                          | Week 10–11 |
| Phase 7 | Testing, deployment, documentation and final presentation          | Week 12    |

### Roles & Contributions

| Student          | Assigned Responsibility                      | Actual Contribution                                                                                                                                  |
|------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Haresh Kumar N L | Model development and system integration     | Developed the Autoencoder-based anomaly detection workflow, supported model evaluation, and integrated the detection results with the web dashboard. |
| Kumaran M        | Data preprocessing and dashboard development | Worked on network-traffic data preprocessing, visualization, dashboard functionality, testing, and documentation.                                    |

### Project Budget

| Item                            | Estimated Cost | Actual Cost |
|---------------------------------|----------------|-------------|
| Dataset / Open-source resources | ₹0             | ₹0          |
| Software and development tools  | ₹0             | ₹0          |
| Cloud / Web deployment          | ₹0–₹1,000      | ₹0          |
| Internet / Data usage           | ₹1,000         | ₹0          |
| Documentation and printing      | ₹500           | ₹0          |
| Total                           | ₹1,500–₹2,500  | ₹0          |

## **CHAPTER 5**

### **CONCLUSION, FUTURE WORK AND REFLECTION**

#### **5.1 Conclusion**

The project successfully addressed the challenge of detecting abnormal network traffic using a Deep Learning-based Autoencoder approach. Network traffic data was collected, cleaned, encoded, normalized, and prepared for model training. The Autoencoder was developed using an Encoder–Bottleneck–Decoder architecture and trained to learn the patterns of normal network traffic. Anomaly detection was performed by calculating reconstruction error and comparing it with a defined threshold to classify traffic as normal or suspicious. The developed system was validated using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix, providing measurable evidence of its detection performance. A web-based dashboard was also implemented to present alerts, monitoring information, and performance results. Overall, the project demonstrates the feasibility of using Autoencoder-based anomaly detection as an intelligent approach for network intrusion detection and provides a functional prototype for network security monitoring.

#### **5.2 Future Work**

The proposed system can be further improved by training the Autoencoder with larger and more diverse network-traffic datasets to improve its generalization capability. The model architecture and anomaly threshold can be optimized to reduce false positives and false negatives. Future development can include advanced Autoencoder variants and hybrid deep learning models for detecting more complex attack patterns. The dashboard can also be enhanced with real-time network traffic collection, continuous monitoring, automated notifications, and detailed threat analysis. Finally, the system can be extended with secure cloud deployment, authentication improvements, and integration with enterprise security monitoring platforms for practical large-scale use.

#### **5.3 Professional Learning & Individual Reflection**

##### **New Knowledge Acquired**

- Learned the principles of Autoencoder-based anomaly detection for network security.
- Gained knowledge of network intrusion datasets and traffic preprocessing techniques.
- Learned feature encoding, normalization, feature selection, and train/test data preparation.
- Understood reconstruction error and threshold-based anomaly classification.
- Learned model evaluation using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.



- Gained practical knowledge of integrating a machine-learning model with a web-based dashboard.
- Learned basic deployment concepts using web hosting and backend API services.

### **Engineering & Professional Skills Developed**

- Deep Learning model development and testing.
- Python programming and data preprocessing.
- Network security and anomaly detection analysis.
- Problem-solving and debugging.
- Web application and API integration.
- Data visualization and technical documentation.
- Team collaboration, task allocation, and project management.

### **Individual Reflection**

The most difficult engineering problem I encountered was integrating the trained Autoencoder with the web-based monitoring system while ensuring that anomaly detection results were displayed correctly. I solved this through systematic debugging, testing the model output and API communication separately before integrating the complete system. A key engineering decision I made was to use reconstruction error with a threshold for anomaly detection because it provides a measurable basis for distinguishing normal and abnormal traffic. I independently learned more about Autoencoders, model evaluation, API integration, and deployment. If the project were repeated, I would improve the model tuning and develop a more robust real-time data-processing pipeline.

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## **APPENDICES**

### **Appendix A – Detailed Calculations**

The Autoencoder evaluates network traffic by calculating the reconstruction error between the original input and the reconstructed output. The Mean Squared Error (MSE) is calculated for each network-traffic sample using:

$$\text{MSE} = (1/n) \sum (x_i - \hat{x}_i)^2$$

where  $x_i$  represents the original input value,  $\hat{x}_i$  represents the reconstructed value, and  $n$  represents the number of features. The calculated reconstruction error is compared with the selected anomaly threshold. If the reconstruction error is greater than the threshold, the traffic is classified as Intrusion/Anomaly; otherwise, it is classified as Normal.

The classification performance is evaluated using the following measures:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

These calculations provide quantitative evidence for assessing the effectiveness of the proposed intrusion detection system.

### **Appendix B – Engineering Drawings / Schematics**

The proposed system architecture consists of four major stages: data preprocessing, Autoencoder model development, anomaly detection and web-based visualization. Network traffic data is first cleaned, encoded and normalized before being provided to the Autoencoder. The Autoencoder consists of an Encoder, Bottleneck and Decoder and learns the patterns of normal traffic. During detection, the reconstructed output is compared with the original input and the reconstruction error is calculated. The error is then compared with the anomaly threshold to classify the traffic as normal or suspicious. The resulting alerts and performance information are displayed through the web-based dashboard.

The system workflow diagrams presented in the report provide the engineering representation of data flow between these components.

## **Appendix C – Source Code / Code Repository Information**

The complete implementation of the proposed Network Intrusion and Anomaly Detection System is maintained in the project repository. The implementation contains the Autoencoder model, data preprocessing procedures, anomaly detection logic, performance evaluation functions, backend API and web-based dashboard components.

The source code is maintained separately from the report to keep the document concise and allow the implementation to be updated independently. The repository also provides the project files required for deployment and further development.

Project Repository: GitHub repository containing the complete project implementation.

## **Appendix D – Datasheets**

Not applicable to this project because the proposed system is a software-based Network Intrusion Detection System and does not contain physical electronic components or manufactured hardware requiring component datasheets.

## **Appendix E – Experimental / Raw Data**

The experimental evaluation uses network-traffic records containing numerical and categorical features required for anomaly detection. The dataset is preprocessed before being supplied to the Autoencoder. Sample outputs include reconstructed traffic values, reconstruction-error values, anomaly classifications and classification-performance results.

The experimental results are used to verify whether the system can distinguish normal traffic from abnormal network activity. The final evaluation includes the confusion matrix together with Accuracy, Precision, Recall and F1-Score.

## **Appendix F – Ethics / Institutional Approval**

No human participants, personal interviews, medical data or experimental subjects were involved in the development and evaluation of this software-based project. The project uses publicly available network-intrusion data for academic and engineering evaluation. Therefore, no separate institutional ethics approval was required for the implementation described in this report.

## **Appendix G – Project Meeting / Progress Records**

The project was developed through progressive stages consisting of problem identification, literature review, requirement analysis, dataset preparation, Autoencoder development, anomaly detection, system integration, testing, deployment and documentation. Team members discussed implementation progress, identified technical issues and divided development and documentation activities according to their assigned responsibilities.

## **Appendix H – User/Industry Feedback**

Formal industry or external-user feedback was not collected as part of this academic prototype. The system was reviewed through functional testing, model evaluation and verification of the implemented dashboard features. Future development can incorporate structured feedback from cybersecurity professionals or network administrators.

## **Appendix I – Publications / Patents / Competitions / Product Outcomes**

No publication, patent or competition outcome was produced as part of the current project. The principal outcome is a functional prototype of a Deep Learning-based Network Intrusion and Anomaly Detection System with a web-based monitoring dashboard.

## **Appendix J – Individual Contribution Evidence**

My contributed to the design, development, testing, analysis, and documentation of the proposed Deep Learning-Based Network Intrusion Detection System. His major contribution included developing the Autoencoder model using the Encoder–Bottleneck–Decoder architecture, implementing reconstruction-error-based anomaly detection, and integrating the trained model with the system workflow. He also contributed to system testing, performance evaluation, troubleshooting, dashboard integration, deployment activities, and preparation of the technical report.

### **Major individual contributions:**

- Designed and implemented the Autoencoder architecture.
- Implemented normal-traffic reconstruction and reconstruction-error calculation.
- Developed the anomaly detection and threshold-based classification logic.
- Contributed to backend and dashboard integration.

- Performed functional testing, debugging, and verification.
- Evaluated system performance using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.
- Contributed to system deployment and GitHub project maintenance.
- Prepared and reviewed technical documentation, diagrams, results, and report sections.